What is Latex?

LaTeX is pronounced as '***lah-tech***' or '***lay-tech***.' which is distributed under the LPPL (Latex Project Public License). It is based on TEX, a typesetting language designed for science and math. Both **LATEX** and **TEX** contain a variety of font styles, such as serif, typewriter, and a set of mathematical functions. It is an open-source and powerful document preparation system. It is high-quality typesetting, which can be used for any publishing. It is often used for scientific publications and medium to large technical documents. It is not a word processor or any word document, but it is an efficient way for the publishers and the authors to organize their materials. It gives a unique appearance to the system. It saves the time for designing as well as the efforts used for the documents.

The **Latex system** handles the layout and formatting of the structures using familiar concepts of the section, table, figure, chapter, etc. It follows the philosophy of separate presentation, i.e., separating the writing contents and the visual appearance so that the authors can focus more on the content rather than its appearance. It is named as the standard for the communication and publication of the technical or scientific documents.

Features of Latex

- Latex is a markup language and free license software. It includes a set of built-in commands.
- It is a mode of mathematical and special symbols.
- LaTeX is generally distributed along with plain **TEX**, i.e., it provides a set of macros for TEX to interpret. The other macros for TEX are Plain TEX, GNU Texinfo, etc.
- LaTeX is not compatible with the GNU (General Public License). It is available on most of the operating systems like UNIX, BSD, Linux, Windows, DOS, etc.

LaTeX/Document Structure

The main point of writing a text is to convey ideas, information, or knowledge to the reader. The reader will understand the text better if these ideas are well-structured, and will see and feel this structure much better if the typographical form reflects the logical and semantic structure of the content.

Given only the logical and semantical structure of a text, LaTeX derives the typographical form of the text according to the “rules” given in the document class file and in various style files. LaTeX allows users to structure their documents with a variety of hierarchical constructs, including chapters, sections, subsections and paragraphs.

Global structure

When LaTeX processes an input file, it expects it to follow a certain structure. Thus every input file must contain the commands

```
\documentclass{...}
\begin{document}
...
\end{document}
```

The area between \documentclass{...} and \begin{document} is called the *preamble*. It normally contains commands that affect the entire document.

After the preamble, the text of your document is enclosed between two commands which identify the beginning and end of the actual document:

```
\begin{document}
...
\end{document}
```

You would put your text where the dots are. The reason for marking off the beginning of your text is that LaTeX allows you to insert extra setup specifications before it (where the blank line is in the example above: we'll be using this soon). The reason for marking off the end of your text is to provide a place for LaTeX to be programmed to do extra stuff automatically at the end of the document, like making an index.

A useful side-effect of marking the end of the document text is that you can store comments or temporary text underneath the \end{document} in the knowledge that LaTeX will never try to typeset them:

```
\end{document}
...
```

Preamble

When processing an input file, LaTeX needs to know which layout standard to use. Layouts standards are contained within 'class files' which have .cls as their filename extension.

```
\documentclass[options]{class}
```

Here, the class parameter for the command \documentclass specifies the .cls file to use for the document. It is recommended to put this declaration at the very beginning. The LaTeX distribution provides additional classes for other layouts, including letters and slides. It is also possible to create your own, as is often done by journal publishers, who simply provide you with their own class file, which tells LaTeX how to format your content. But we'll be happy with the standard article class for now. The options parameter customizes the behavior of the document class. The options have to be separated by commas.

Example: an input file for a LaTeX document could start with the line

```
\documentclass[11pt,twoside,a4paper]{article}
```

which instructs LaTeX to typeset the document as an article with a base font size of 11 points, and to produce a layout suitable for double sided printing on A4 paper.

Here are some document classes that can be used with LaTeX:

Document Classes

article	For articles in scientific journals, presentations, short reports, program documentation, invitations, ...
IEEEtran	For articles with the IEEE Transactions format.
proc	A class for proceedings based on the article class.
minimal	It is as small as it can get. It only sets a page size and a base font. It is mainly used for debugging purposes.
report	For longer reports containing several chapters, small books, thesis, ...
book	For books.
slides	For slides. The class uses big sans serif letters.
memoir	For sensibly changing the output of the document. It is based on the book class, but you can create any kind of document with it [1]
letter	For writing letters.
beamer	For writing presentations

The generic document classes that come with LaTeX offer some layout flexibility, which is why they have a lot of options in common. Non-generic classes (those provided by university departments or publication houses) may have different options than those shown below or no options at all. Normally, third-party classes come with their own documentation. The most common options for the generic document classes are listed in the following table:

Document Class Options

10pt, 11pt, 12pt	Sets the size of the main font in the document. If no option is specified, 10pt is assumed.
a4paper, letterpaper,...	Defines the paper size. The default size is letterpaper; However, many European distributions of TeX now come pre-set for A4, not Letter, and this is also true of all distributions of pdfLaTeX. Besides that, a5paper, b5paper, executivepaper, and legalpaper can be specified.
fleqn	Typesets displayed formulas left-aligned instead of centered.
leqno	Places the numbering of formulas on the left hand side instead of the right.
titlepage, notitlepage	Specifies whether a new page should be started after the document title or not. The article class does not start a new page by default, while report and book do.
twocolumn	Instructs LaTeX to typeset the document in two columns instead of one.
twoside, oneside	Specifies whether double or single sided output should be generated. The classes article and report are single sided and the book class is double sided by

	default. Note that this option concerns the style of the document only. The option twoside does not tell the printer you use that it should actually make a two-sided printout.
landscape	Changes the layout of the document to print in landscape mode.
openright, openany	Makes chapters begin either only on right hand pages or on the next page available. This does not work with the article class, as it does not know about chapters. The report class by default starts chapters on the next page available and the book class starts them on right hand pages.
draft, final	final is default. draft makes LaTeX indicate hyphenation and justification problems with a small square in the right-hand margin of the problem line so they can be located quickly by a human. It also suppresses the inclusion of images and shows only a frame where they would normally occur.

For example, if you want a report to be in 12pt type on A4, but printed one-sided in draft mode, you would use:

```
\documentclass[12pt,a4paper,oneside,draft]{report}
```

Packages

While writing your document, you will probably find that there are some areas where basic LaTeX cannot solve your problem. If you want to include graphics, colored text or source code from a file into your document, you need to enhance the capabilities of LaTeX. Such enhancements are called packages. Some packages come with the LaTeX base distribution. Others are provided separately. Modern TeX distributions come with a large number of packages pre-installed. The command to use a package is pretty simple: \usepackage:

```
\usepackage[options]{package}
```

command, where package is the name of the package and options is a list of keywords that trigger special features in the package. For example, to use the color package, which lets you typeset in colors, you would type:

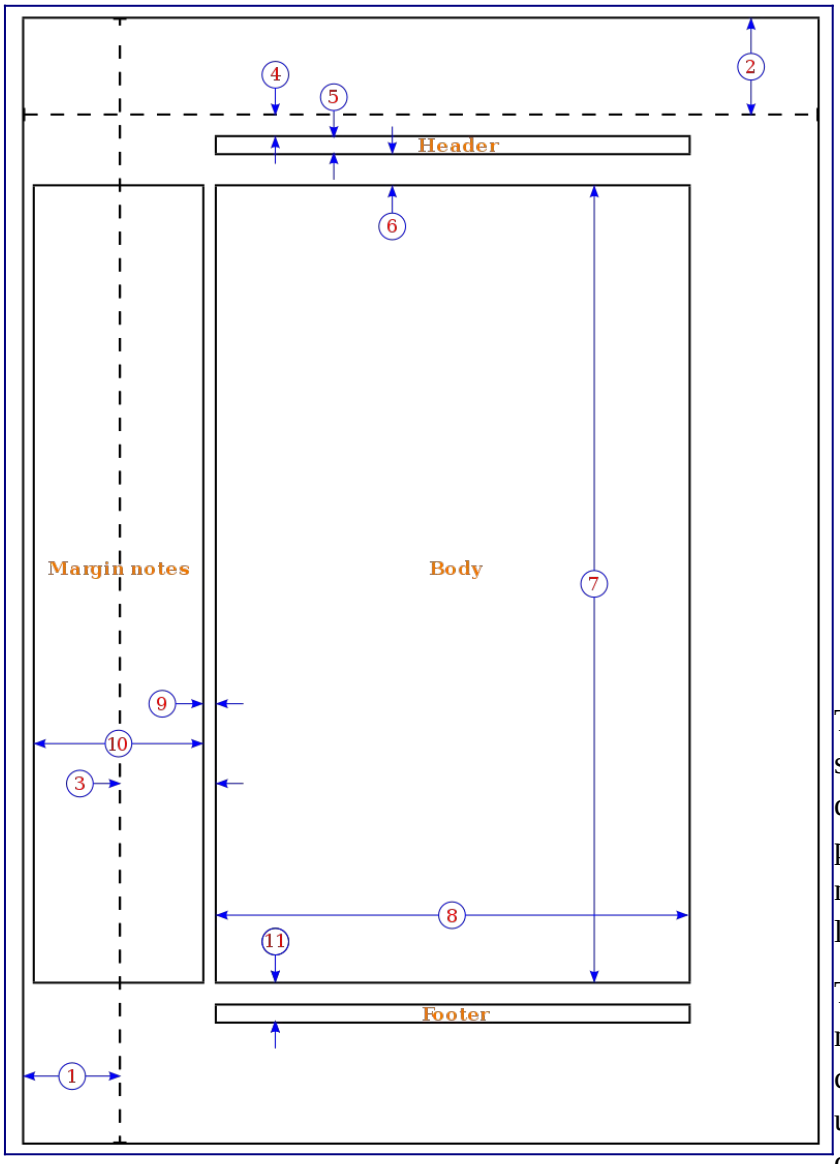
```
\documentclass{report}
\usepackage{color}

\begin{document}
...
\end{document}
```

You can pass several options to a package, each separated by a comma.

```
\usepackage[option1,option2,option3]{"package_name"}
```

Page dimensions

A page in LaTeX is defined by many internal parameters. Each parameter corresponds to the length of an element of the page, for example, \paperheight is the physical height of the page. Here you can see a diagram showing all the variables defining the page. All sizes are given in TeX points (pt), there are 72.27pt in an inch or 1pt ≈ 0.3515mm.



1. one inch + \hoffset
2. one inch + \voffset
3. \oddsidemargin = 31pt
4. \topmargin = 20pt
5. \headheight = 12pt
6. \headsep = 25pt
7. \textheight = 592pt
8. \textwidth = 390pt
9. \marginparsep = 10pt
10. \marginparwidth = 35pt
11. \footskip = 30pt

- \marginparpush = 7pt (not shown)
- \hoffset = 0pt
- \voffset = 0pt
- \paperwidth = 597pt
- \paperheight = 845pt

The current details plus the layout shape can be printed from a LaTeX document itself. Use the *layout* package and the command of the same name: \usepackage{layout} ... \layout{}

To render a frame marking the margins of a document you are currently working on, add \usepackage{showframe}to the document.

Margins

The commands are used to set the margins manually. The margins internally are considered **1 inch** by default. Hence, if you want to set a margin of **0.5** inches, you have to mention that margin **-0.5** inches. You can also set the height and width of the text area by using the **textheight** and **textwidth** commands.

The available commands to set margins are given below in the table:

Margins	command
left margin	**\oddsidemargin**

(for odd pages if you are using single-sided)	
left margin (for the even pages if you are using double-sided)	**\evensidemargin**
right margin	**\textwidth**
top margin	**\topmargin**
bottom margin	**\texthe**

The commands used to set the inches for margins are:

- To set the top margin 0.7 inch, **\setlength{\topmargin}{-0.7in}**
- To set margin 1 inch on right pages, **\setlength{\oddsidemargin}{0in}**
- To set margin 1 inch on left pages in two-sided document, **\setlength{\evensidemargin}{0in}**
- To leave the space for right margin, **\setlength{\textwidth}{0in}**
- To reserve 9 inches for the text , **\setlength{\textheight}{9in}**

Headers and Footers

Appears in the head or the foot sections. The output screen consists of a head, body, and foot. The page number is printed by default by all the classes except the letter.

To modify the default actions, you can use the **\pagestyle** command, which is placed just after the **\chapter** or the command similar to this.

The types of standard page styles are given below:

- **\pagestyle{plain}**
  It is the default style for the report classes and the articles. The page number is listed on the foot while the head is empty.
- **\pagestyle{headings}**
  In this style, the foot is empty. The section heading and the page number is put in the head.
- **\pagestyle{myheadings}**
  To specify the information used in the head; the following commands are used.

1. \markboth{leftheader}{rightheader}
2. \markright{rightheader}

\markboth is used for a two-sided file or document, while **\markright** is used for a one-sided document.

- **\pagestyle{empty}**
  In this style, both the header and foot are empty.
- **\thispagestyle{style}**
  It is used to change the style for the current page only. For example, **\thispagestyle{empty}** is used to have nothing in the foot and head. This command will be executed only for the current page without affecting other pages.

If you want to format or customize your header and footers in another way. You can use the **fancyhdr** command. This command in Latex is used as **\usepackage{fancyhdr}**. The package used

here can align your header and footers to the left, right, and the center. You can also define multi-line headers and filters, separate headers used for the even and odd pages, etc.

\pagestyle{fancy} is also used here.

The following commands can be included in the {} (curly brackets): **\lhead{}, \chead{}, \rhead{}, \lfoot{}, \cfoot{}, \rfoot{}**.

If you want to suppress the horizontal line under the header, which is drawn by default, you can use the **\renewcommand{\headrulewidth}{0pt}** command.

`fancyhdr` is used to set header and footer. We take care of using the `fancy` style and to start from scratch by erasing the previous header and footer with `\fancyhf{}`.

```
%%%
```

Notice that you do not need to specify section numbers; LaTeX will sort that out for you. Also, for sections, you do not need to use \begin and \end commands to indicate which content belongs to a given block.

LaTeX provides 7 levels of depth for defining sections (see table below). Each section in this table is a subsection of the one above it.

Command	**Level**	**Comment**
\part{"part"}	-1	not in letters
\chapter{"chapter"}	0	only books and reports
\section{"section"}	1	not in letters
\subsection{"subsection"}	2	not in letters
\subsubsection{"subsubsection"}	3	not in letters
\paragraph{"paragraph"}	4	not in letters
\subparagraph{"subparagraph"}	5	not in letters

All the titles of the sections are added automatically to the table of contents (if you decide to insert one). But if you make manual styling changes to your heading, for example a very long title, or some special line-breaks or unusual font-play, this would appear in the Table of Contents as well, which you almost certainly don't want. LaTeX allows you to give an optional extra version of the heading text which only gets used in the Table of Contents and any running heads, if they are in effect. This optional alternative heading goes in [square brackets] before the curly braces:

```
\section[Effect on staff turnover]{An analysis of the
effect of the revised recruitment policies on staff
turnover at divisional headquarters}
```

LATEX LISTS

The Latex lists are used to add any additional packages. There are three types of lists. One is the **enumerate** list, which produces the numbered list. The second is **itemize**, which is used for the bulleted lists. The third one is the **description list**, which is neither bulleted nor numbered. In Latex, each list is defined by the **\item**. Through this method, sub-lists can also be created.

The example for the lists is shown below:

```
\begin{enumerate}
\item first point
\item second point
\begin{itemize}
\item first sub-point
\item second sub-point
\end{itemize}
\item third point
\end{enumerate}
```

1. first point
2. second point
   - first sub-point
   - second sub-point
3. third point

1.Enumerate

It is used to create numbered lists.

The simple code for the enumerate list is given below:

```
\documentclass{article}
\begin{document}
\begin{enumerate}
\item this is the first point
\item this is the second point
\end{enumerate}
\end{document}
```

1. this is the first point
2. this is the second point

2. Itemize

It is used to create the unnumbered list. The example is given below:

```
\documentclass{article}
\begin{document}
\begin{itemize}
\item this is the first point
\item this is the second point
\end{itemize}
\end{document}
```

- this is the first point
- this is the second point

3. Description

The description list is least known. It is used when you need to explain terms or notations. The program or code for the Description list is given below:

```
\documentclass{article}
\begin{document}
\begin{description}
\item[Chemistry] the study of behavior of elements and compounds
\item[History] the study of past
\item[Geology] the study of the solid Earth
\end{description}
\end{document}
```

Chemistry the study of behavior of elements and compounds

History the study of past

Geology the study of the solid Earth

Nested Lists

For all the above lists, **Nested Lists** are used to create different environments. It also means that it is possible to have a sub-list for an item of a list. The maximum depth for the number of levels is four. The example for the nested list is given below:

```
\documentclass[12pt]{article}
\begin{document}
\begin{enumerate}
\item One
\begin{enumerate}
\item First
\item Second
\item Third
\end{enumerate}
\item Two
\item Three
\end{enumerate}
\end{document}
```

1. One
   (a) First
   (b) Second
   (c) Third
2. Two
3. Three

The numbering styles of the nested lists depends on the depth of the lists. Let-s consider another example:

```
\documentclass[17pt]{article}
\begin{document}
\begin{enumerate}
\item first is entered
\item another item of first
\begin{enumerate}
\item Second one
\item another item of second
\begin{enumerate}
\item third is entered
\item another item of third
\begin{enumerate}
\item Fourth one
\item another item of fourth
\end{enumerate}
\end{enumerate}
\end{enumerate}
\end{enumerate}
\end{document}
```

1. first is entered
2. another item of first
   (a) Second one
   (b) another item of second
      i. third is entered
      ii. another item of third
         A. Fourth one
         B. another item of fourth

➔ Latex lists are also categorized depending on the order, namely, **ordered list** and **unordered lists**. The example of both these lists is given below:

1. Ordered Lists

```
\documentclass[12pt]{article}
\begin{document}
\begin{enumerate}
\item First
\item Second
\item Third
\end{enumerate}
\end{document}
```

Latex provides the **enumerate** environment for the ordered lists. Consider an example.

1. First
2. Second
3. Third

2. Unordered Lists

For Unordered lists, Latex provides the **itemize** environment. It also works without any additional packages. In the example given below, the code is the same as that of an ordered list, just in place of **enumerate, itemize** is used.

```
\documentclass[12pt]{article}
\begin{document}
\begin{itemize}
\item First
\item Second
\item Third
\end{itemize}
\end{document}
```

- First
- Second
- Third

For both the environments, Latex provides the **\item** command, which has to be declared in the beginning.

Changing the bullet and numbering pattern/format of Latex Lists

Sometimes, it becomes necessary to change the bullets and numbering of the lists to use a different format. As we know, for the input of a special character, $ sign is used at the beginning and at the end. The method for both the ordered and unordered list is given below:

- **Unordered lists**

If you change the format from bullets to dash, then the following command is used: `\item[--]` or `\item[$-$]`

If you change the format from bullets to asterisk, then the following command is used: `\item[$\ast$]`

If you change the format from bullets to particular character, then the following command is used: `\item[$ \Any character$]`

The program to change the bullets is given below:

```
\documentclass[14pt]{article}
\begin{document}
\begin{itemize}
\item[--] Dash
\item[$-$] Dash
\item[$\ast$] Asterisk
\item[$\#$] hash character used
\end{itemize}
\end{document}
```

– Dash

– Dash

∗ Asterisk

hash character used

- **Ordered lists**

It is a tricky process if you want to change the environment. The easiest step is to use the **enumitem** environment. This is used with the usepackage command like **usepackage{enumitem}**.

```
\documentclass[12pt]{article}
\usepackage{enumitem}
\begin{document}
\begin{enumerate}[label=(\alph*)]
\item one item
\item two item
\begin{enumerate}[label=(\arabic*)]
\item third item
\item fourth item
\begin{enumerate}[label=(\roman*)]
\item fifth item
\item sixth item
\end{enumerate}
\end{enumerate}
\end{enumerate}
\end{document}
```

(a) one item

(b) two item

(1) third item

(2) fourth item

(i) fivth item

(ii) sixth item

➔ You can also use **enumitem** option for unordered list to change the symbol for all items of the list.

```
\documentclass{article}
\begin{document}
\renewcommand{\labelitemi}{$\ast$}
\renewcommand{\labelitemii}{$\cdot$}
\renewcommand{\labelitemiii}{$\diamond$}
\begin{itemize}
\item this is the first point
\begin{itemize}
\item this is the second point
\begin{itemize}
\item this is the third point
\end{itemize}
\end{itemize}
\end{itemize}
\end{document}
```

- **Itemize**

In this type of lists, the renewcommand is used to change the bullets size at different levels. Let-s understand this by an example. The commands can be used for the formats to replace the bullet points.

The output is shown in the below image:

∗ this is the first point

· this is the second point

⋄ this is the third point

➔ The label commands used at the different levels are:

- labelitemi for level 1
- labelitemii for level 2
- labelitemiii for level 3
- labelitemiv for level 4

- **Numbered list styles**

The styles available for the numbered lists are given below:

Code	**Description**
\Alph	Used for the uppercase (A, B, C...)
\alph	Used for the lowercase (a, b, c, d....)
\roman	For the lowercase roman numerals (i, ii, iii, iv...)
\Roman	For the uppercase roman numerals (I, II, III, IV...)
\arabic	Used for the Arabic numbers (1, 2, 3, 4, 5...)

To change the start number or starting you can use the **\setcounter** for the enumerate type lists. The code for this type is shown in the below example:

```
\documentclass{article}
\begin{document}
\renewcommand{\labelitemiii}{\Roman{enumiii}}
\begin{enumerate}
\item first item
\item first number item
\begin{enumerate}
\item second item
\item second number item
\begin{enumerate}
\setcounter{enumiii}{6}
\item third item
\item third number item
\end{enumerate}
\end{enumerate}
\end{enumerate}
\end{document}
```

The output is shown in the image given below:

1. first item
2. first number item
   (a) second item
   (b) second number item
      vii. third item
      viii. third number item

Generate a table of contents in LaTeX

```
\
documentclass{article}

\begin{document}

\tableofcontents
\newpage

\section{Section}

Dummy text

\
subsection{Subsection}

Dummy text

\end{document}
```

LaTeX offers features to automatically generate a table of contents, a list of figures and a list of tables.

Generating a *table of contents* can be done with a few simple commands. LaTeX will use the section headings to create the *table of contents* and there are commands to create a *list of figures* and a *list of tables* as well. I will give a small example code to create a *table of contents* first:

Contents

1 Section **2**

1.1 Subsection 2

Depth

Sometimes it makes sense to only show a subset of the headings for all sections or for a particular section. For this reason you can set a the *tocdepth*by using the command \setcounter{tocdepth}{X}, where X is the desired depth. A value of 0 means that your table of contents will show nothing at all and 5 means, that even subparagraphs will be shown. The value has to be set in the preamble of your document and automatically applies to the whole document:

```
% ...

\setcounter{tocdepth}{1} %
Show sections
%\setcounter{tocdepth}{2} % +
subsections
%\setcounter{tocdepth}{3} % +
subsubsections
%\setcounter{tocdepth}{4} % +
paragraphs
%\setcounter{tocdepth}{5} % +
subparagraphs

\begin{document}
%...
\tableofcontents
%...
\end{document}
```

Contents

1 Section **2**

If you don't want to change the depth for all sections, you can also adjust the tocdepth for each section individually. In this case you don't have to set the tocdepth before the section which should have more or less depth.

```
%...

\begin{document}
%...
\addtocontents{toc}{\setcounter{tocdepth}{1}} % Set depth to 1
\section{Another section}
\subsection{Subsection}
\subsubsection{Subsubsection}
%...
\addtocontents{toc}{\setcounter{tocdepth}{3}} % Reset to default (3)
\end{document}
```

This will generate the following table of contents, using the default tocdepth for the first section, but tocdepth = 1 for this section:

1 Section **2**

2 Another section **2**

2.1 Subsection 2

2.1.1 Subsubsection 2

Spacing

If you're not happy with the spacing of the headings in your table of content, the easiest way of changing the spacing of your table of contents (and document in general) is by using the setspace package. First add *\usepackage{setspace}* to your preamble:

```
%...

\usepackage{setspace}
%...
\begin{document}
%...
```

You can then proceed to set the spacing for individual parts of your document, including the table of contents like so:

```
%...

\begin{document}
%...
\doublespacing
\tableofcontents
\singlespacing
%...
```

Which will lead to the following output:

Contents

1 Section **2**

1.1 Subsection 2

2 Another section **3**

Latex Images

The images are used to enhance the information in a document or in any file. In most of the scientific documents, pictures are essential. In Latex, you can enlarge, shrink, rotate, and can set the reference of the image in your document.

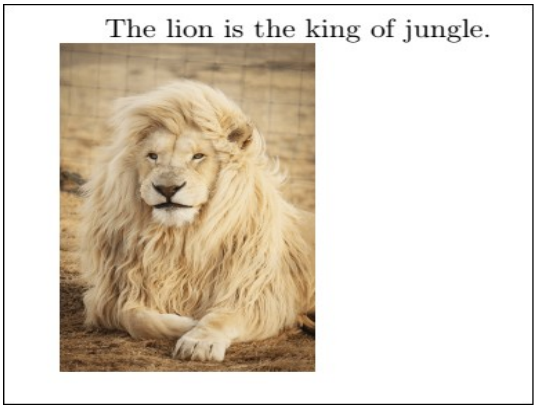
Latex does not insert the picture itself.

The basic requirements to insert an image are:

- Including the **graphicx** package. The command will be written as **\usepackage{graphicx}**.
- You need to download the particular image from the browser and save that image in the same folder where the Latex files are present. Otherwise, Latex cannot access those images. Mention the same in the code or program that you have mentioned while saving that particular image.

➔ Let's consider a simple example to insert an image. The code is written below:

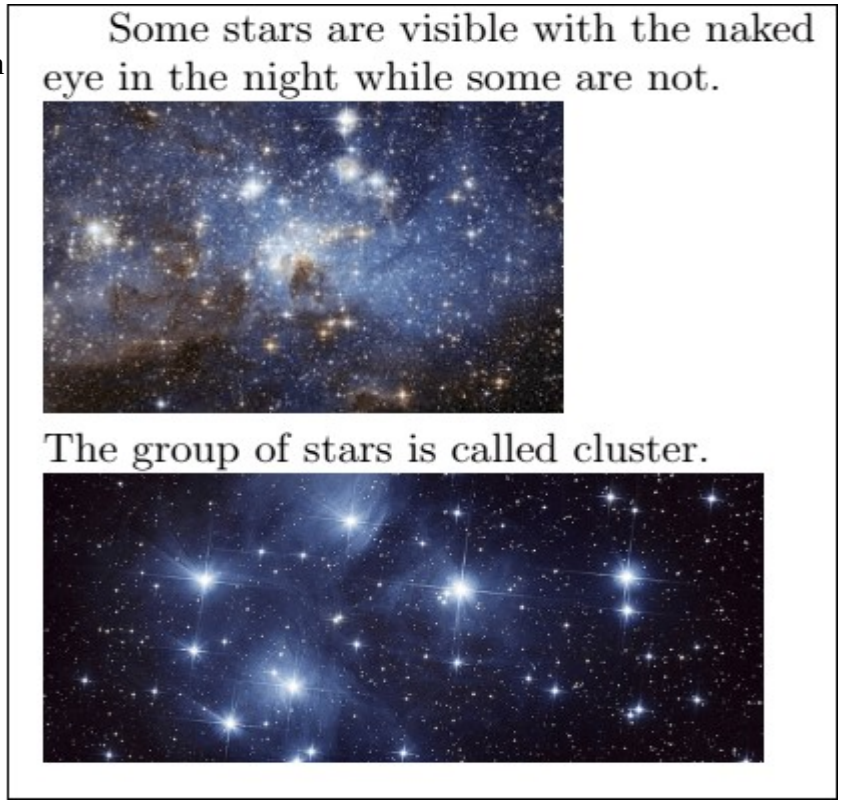
```
\documentclass[12pt]{article}
\usepackage{graphicx}
\begin{document}

The lion is the king of jungle.\\
\includegraphics[scale=0.1]{lion}

\end{document}
```



➔ You can insert many pictures in your document, depending on the requirements.

The code to insert two images is given below:

```
\documentclass[12pt]{article}

\usepackage{graphicx}
\begin{document}
Some stars are visible with the naked \\ eye
in the night while some are not.\\
\includegraphics[scale=0.5]
{star}\\ % star and star1 are the images save
d with these names.
The group of stars is called cluster.\\
\includegraphics[scale=0.25]{star1}
\end{document}
```



Path Specification

Sometimes, there are a large number of images present on your computer in different folders. To insert those images directly in your Latex document, you can specify the path using the \ **graphicspath**{{ path}} command.

Let's consider the **path-images**. The command will be written as **\graphicspath{{ \images}}**. You need to create the folder by the name 'images' so that Latex can easily insert images or pictures into the document through this path.

The example is given below:

```
\documentclass[12pt]{article}

\usepackage{graphicx}
\begin{document}
\graphicspath{ {images/} }
\includegraphics[scale=0.65]
{cloud}\\
The image of the cloud is given a
bove.
\end{document}
```

The image of the cloud is given above.

➔ You can also specify the absolute path if the exact location of the file is specified in your system. The example of such type is given below:

```
\documentclass[12pt]{article}

\usepackage{graphicx}
\begin{document}

\graphicspath{ {c:/images1/} }
\includegraphics[scale=0.35]{butterfly}\\
The image of the butterfly is given above
.
\end{document}
```

The image of the butterfly is given above.

Here, we have created the 'images1' folder in the C: drive and hence specified the path. Similarly, you can also specify the path even if the image is present in the subfolder.

Changing the size of Image

You can specify the height and width of the image according to the requirements. The example is given below:

```
\documentclass[12pt]{article}
\usepackage{graphicx}
```

```
\begin{document}

\graphicspath{ {images/} }
\includegraphics[width=6cm, height=4cm]
{smiley}\\
The image of a smiley is given above.

\end{document}
```

The image of a smiley is given above.

Here, you can use any path and dimensions according to the requirements.

Rotation of Image

You can rotate any image by specifying the angle in the **\includegraphics** command. Let's consider a single picture of a butterfly.

The code for such type is given in the below example:

```
\documentclass[12pt]{article}
\usepackage{graphicx}
\begin{document}

\graphicspath{ {images/} }
\includegraphics[scale=0.2, angle=50]
{butterfly}\\
The rotated image is shown above.

\end{document}
```

The rotated image is shown above.

Image alignment

In the image alignment, two commands are used. The first command is **\baselineskip**, which is used to generate vertical line spacing. The second command is the **\quad**, which is used to maintain the horizontal space. It has only a width.

If you want to align the images in the matrix form, then make sure that the width mentioned of all the photos is equal.

The code to arrange the six images in matrix form is given below:

```
\documentclass{article}
\usepackage{graphicx}

\begin{document}
\begin{center}
\graphicspath{ {images/} }
\includegraphics[width=.2\linewidth]{smiley}\quad
\includegraphics[width=.2\linewidth]{smiley}\quad
```

```
\includegraphics[width=.2\
linewidth]{smiley}
\\[\baselineskip]
% adds vertical line spacing
\includegraphics[width=.2\
linewidth]{cloud}\quad
\includegraphics[width=.2\
linewidth]{smiley}\quad
\includegraphics[width=.2\
linewidth]{cloud}
\end{center}
\end{document}
```